



# **M7XC**

## **Compact Satellite Modem Manual**

### **Appendix D: ACM Operation**



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NOTE: All appendices may not be present in manual. Some appendixes may be shipped with the option.

## 1 **Preface**

This Appendix is to be used in conjunction with the main M7XC manual. The manual and appendices are intended for use by personnel with reasonable knowledge of satellite communications and earth station terminology. This manual provides Installation, Operating, and Maintenance procedures for the M7XC Compact Satellite Modem and available options.

### **Revision History**

| Release Date | Comments               |
|--------------|------------------------|
| 06/16/2024   | Initial Public Release |



## 2 Introduction

### 2.1 ACM (Adaptive Coding and Modulation)

ACM (Adaptive Coding and Modulation) is a communication technology that adapts the transmitted MODCOD (modulation and coding) to maximize user throughput for a given link condition.

As the link conditions improve a higher order MODCOD is used so that more user bits of information are transmitted over a (typically fixed) power and bandwidth and fewer error correction bits are transmitted. This maximizes user throughput.

As conditions worsen the modem will use a lower order MODCOD (more robust coding) selection which lowers the user throughput but provides more resiliency to the link availability.

ACM is an adaptive and dynamic algorithm that is constantly maximizing the user throughput over a given noisy channel. The algorithm's function is to maximize user traffic over a fixed BW link. When running ACM, the link will experience more or less user traffic and conversely less or more coding protection to ensure that the link remains error free. If ACM is enabled and link conditions trigger changes to MODCODs, the user traffic will experience changes (increase or decrease) in user throughput.

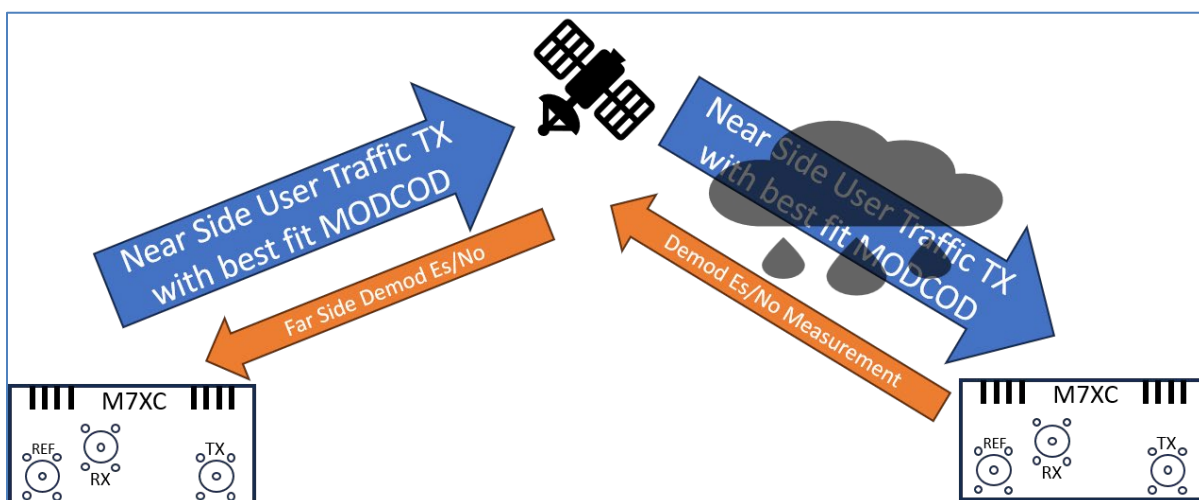


Figure 1-1 ACM Communications

ACM functions by sending link quality measurements from the (far side) receiving demodulator to the near side transmitting modulator over the modem's embedded in-band control channel (MCC channel). The near side modulator uses the link quality measurements (as measured by the far side demodulator) plus any manually entered link margin to select the best fit MODCOD to use for the user traffic transmission. The modulator will use lower order MODCODs when the link requires more protection from noise and interference and higher order MODCODs when the link is more robust or clean. The ACM messages are sent from the demodulator to

the associated modulator at ~100ms intervals. (See Figure 1-1 ACM Communications).

### 3 Operation of ACM

#### 3.1 ACM GUI Interface Settings

There are many user parameters that can be configured to maximize the effect of the ACM algorithm and user experience. Most of these settings can be found on the GUI interface Tx/Modulator page seen below in Figure 2-1 ACM Settings in TX / Modulator GUI.

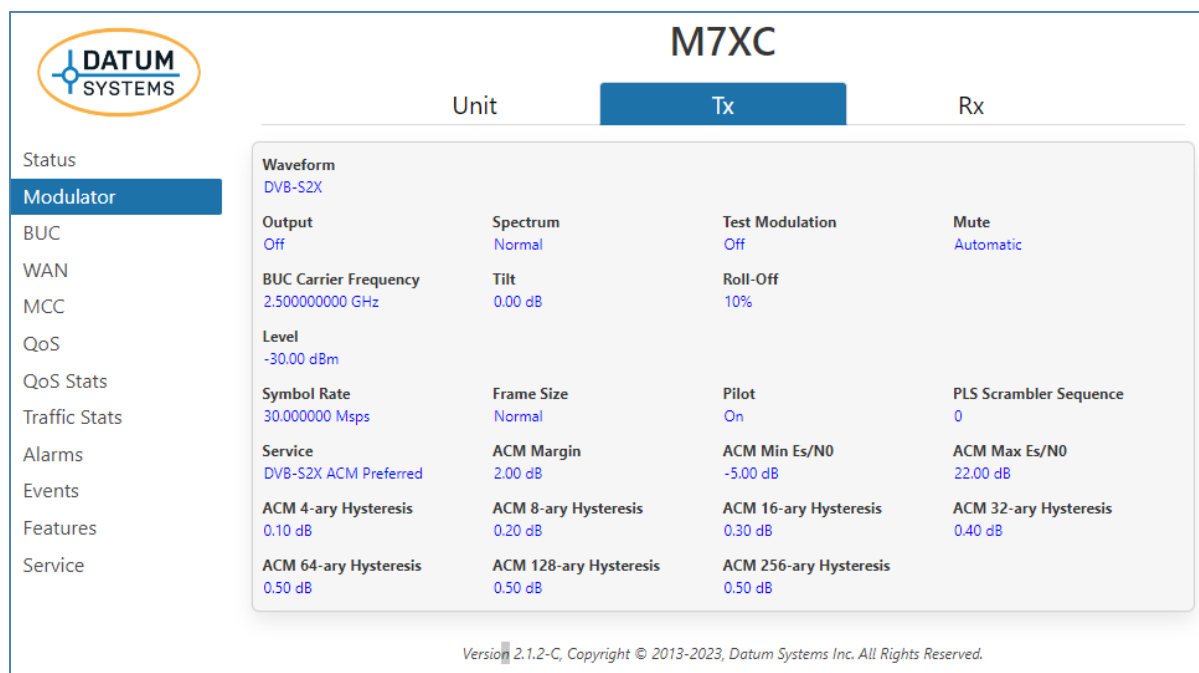


Figure 2-1 ACM Settings in TX / Modulator GUI

##### 3.1.1 Waveform (selection)

In order to run ACM, a “Waveform” setting that supports ACM must be selected. As of the writing of this Appendix the ACM supported waveforms are “LDPC ACM” and “DVB-S2X”.

The “Waveform” selection enables the primary waveform and set(s) of MODCODS that are available within the waveform family that the ACM algorithm can choose from.

##### 3.1.2 Service (selection)

In order to run ACM a “Service” setting that supports ACM must be selected.

The “Service” selection is the subset of MODCODs within the “Waveform” selection that the ACM algorithm can choose from. In LDPC ACM there is only one service option “FlexLDPC ACM”. In DVB-S2 / S2X the options are the following:

- DVB-S2 ACM Preferred: For use with systems that only use DVB-S2 (non DVB-S2X) MODCODS.
- DVB-S2X ACM Preferred: Default selection with the widest range of all DVB-S2X MODCODS to choose from assuming typical (non-near-linear) link conditions.
- DVB-S2X ACM Preferred-L: For use with highly linear (very-low AM-AM and AM-PM) non distorted link conditions.

### 3.2 ACM Margin (setting)

The ACM Margin essentially “modifies” or increases the QEF Es/No point for ALL MODCODs in the available MODCOD pool. By effectively increasing the QEF MODCOD switching point the user has provided a margin of insurance that the ACM algorithm will only switch to MODCODs when the required Es/No + ACM Margin is met making the switch more robust and less prone to errors.

ACM Margin is a VERY IMPORTANT setting as the modem’s specified QEF Es/No point per MODCOD are measured in nearly a perfect AWGN environment and there is little real world messaging delay in a bench test environment.

In a real-world environment, a site located in a 3dB per second rain fade, ACM Margin would be required to keep the link running without errors.

Example 1: Assume a remote site is gathering Es/No information every 100ms on over a typical GEO satellite. Immediately after an Es/No measurement is taken, a 3dB/second rain fade conditions begins at the remote location. This means the next measurement is 100ms after the fade has begun. There is ~50ms to process the message, the one-way satellite link to the hub side modulator is 250ms, it takes the hub modulator 50ms to process the new Es/No measurement and another 250ms to return the newly coded signal to the far side demod.

In total, with a standard GEO environment the RX signal coding is  $100+50+250+50+250 = 700\text{ms}$  behind a fade condition. At 3dB per second of maximum fade the signal could be 2.1dB worse than last reported ( $3\text{dB/s} * 0.7 \text{ sec} = 2.1\text{dB}$ ). In this example, the user would want to put a minimum of 2.1dB of ACM Margin into the system:

Table 2-1 ACM Margin Lookup

|                          | MIN ACM MARGIN: Modem Messaging Interval (100ms) + Modem Processing (100ms) |                         |                        |
|--------------------------|-----------------------------------------------------------------------------|-------------------------|------------------------|
| Maximum Fade Cond (dB/s) | + GEO RTT Delay (500ms)                                                     | + MEO RTT Delay (150ms) | + LEO RTT Delay (40ms) |
| 1 dB/s                   | 0.70 dB                                                                     | 0.25 dB                 | 0.14 dB                |
| 2 dB/s                   | 1.40 dB                                                                     | 0.50 dB                 | 0.28 dB                |
| 3 dB/s                   | 2.10 dB                                                                     | 0.75 dB                 | 0.42 dB                |
| 4 dB/s                   | 2.80 dB                                                                     | 1.00 dB                 | 0.56 dB                |
| 5 dB/s                   | 3.50 dB                                                                     | 1.25 dB                 | 0.70 dB                |
| 6 dB/s                   | 4.20 dB                                                                     | 1.50 dB                 | 0.84 dB                |

Example: DVB-S2X 8PSK 3/5 MODCOD specification requires an Es/No of 5.50 dB to lock without errors. Entering an ACM Margin of “2.10 dB” tells the transmitting modem that the selection of 8PSK 3/5 as the MODCOD will only be allowed if the far side demodulator is reporting an Es/No of 7.60 dB or higher (normative 5.50 + 2.1dB = 7.60dB).

Note: How much margin needed depends on how fast signal fades occur and the speed of the messaging time. The ACM measured Es/No values are averaged / processed and sent every ~100ms. These messages are delayed due to satellite link and processing delay. More margin may be needed if signal fades occur more rapidly; allowing more time for the ACM to catch up with those changes. See Table 2-1 ACM Margin Lookup for examples on margin figures to use.

### 3.3 ACM Min Es/No (setting)

This is the Es/No value the near side modulator will ASSUME of the far side demod when 1) establishing a link for the first time 2) after far side demod unlock >0.5 seconds or 3) when the near side TX modulator has not seen Es/No messages for >1.0 second.

This will essentially establish the minimum MODCOD transmitted by the modulator for initial lock. Once the far end is locked and messages resume, the Tx ACM MODCODs can track above and below this value to maintain lock. Ideally this should be set to the lowest expected operating Es/No value.

To remove this limitation set the ACM Min Es/No to -5.0 dB. This will ensure the link stays up as much as possible but will allow user throughputs that may be under contracted CIR values.

### 3.4 ACM Max Es/No (setting)

Set the "ACM Max Es/No" to establish an upper limit of Tx MODCOD selection used. If the link performs poorly at some high Es/No MODCOD due to channel distortion, phase noise, etc. this can be used to prevent those MODCODs from being selected.

### 3.5 ACM \*-ary Hysteresis (setting)

Set the "ACM \*-ary Hysteresis" to add additional margin before switching “up” to a higher order MODCOD within that \*-ary family. This value is added to the ACM

Margin and QEF MODCOD Es/No value as the threshold to enter that modulation order MODCOD. Once in that MODCOD only the QEF and margin are used for that MODCOD's lower threshold to remain in it.

This can be used to give added margin or assurance when switching into very high order MODCODs such as 32APSK, 64APSK or higher to ensure that the link is linear and clean enough to manage these higher order modulation types.

### **3.6 ACM MODCOD Table (In Operation)**

The actual MODCODs that are in operation when ACM is enabled is a function of the Waveform selected, the Service selected, ACM Margin, Min and Max Es/No as well as any ACM -ary Hysteresis settings.

Taking all of these factors into consideration, the modem will then “prune” or remove any MODCODs that require an Es/No that is higher than another MODCOD with the same or higher spectral efficiency. In other words, MODCODs that (after adjustments) do not have a system link benefit are removed. For this reason, there is no single table of ACM MODCODs but rather a “resulting” table based on the above-mentioned user criteria.

Example 1: If the Waveform selected was DVB-S2X and the Service selected was “DVB-S2X ACM Preferred” the ACM MODCOD selection algorithm would choose non -L (Non-Linear) MODCODs over more efficient -L Linear MODCODs as the system can not depend on a near Linear link.

Example 2: The ACM 32APSK hysteresis is set to 3dB while the QPSK, 8PSK and 16APSK ACM hysteresis settings are 0.0dB. This will tell the modem essentially NOT to switch to 32APSK unless there is sufficient margin due to known non-linearity of a link. Some lesser efficient 16APSK MODCODS that would normally be “pruned” are now used because the 32APSK modulation selections are further out of reach.